Solutions to Mathematical Analysis Problems

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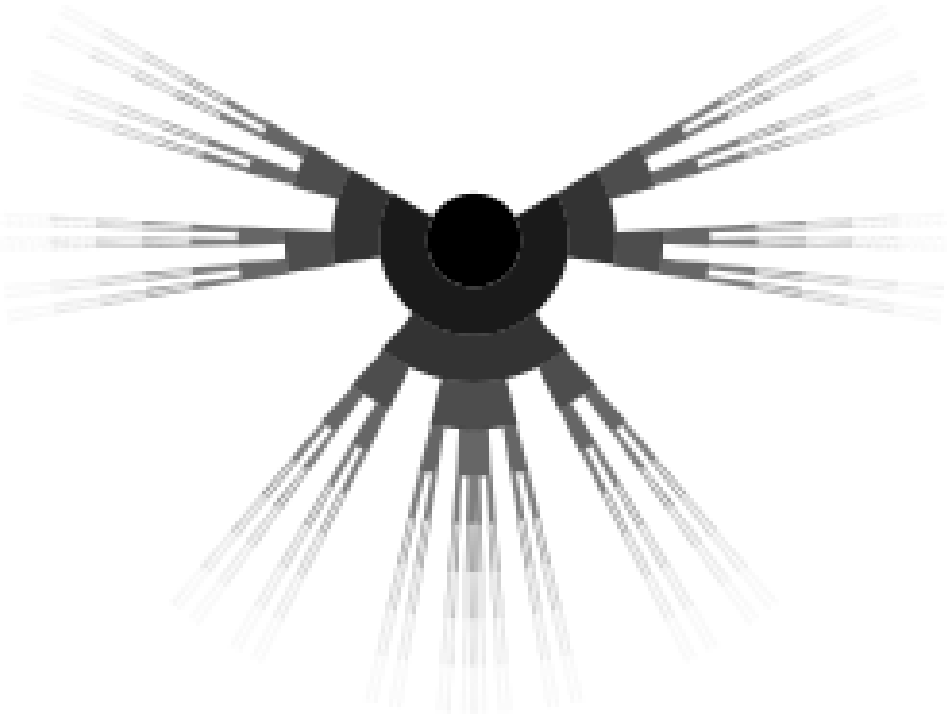
Abstract

This note collects my solutions to selected problems in mathematical analysis. The goal is to present clear arguments and organized solutions for study and review. Books that I use for assignments sources include:

- Real Mathematical Analysis 2e, Charles Pugh
- Understanding Analysis 2e, Stephen Abbott



Problem 1.

Let $\mathcal{T}$ be the collection of open subsets of a metric space M, and $\mathcal{K}$ the collection of closed subsets. Show that there is a bijection from $\mathcal{T}$ onto $\mathcal{K}$.

Source: Real Mathematical Analysis, Pugh, Ex.29 p.127

Solution:

Define $f : \mathcal{T} \to \mathcal{K}$ by $f(A) = A^c$. Since the complement of an open set is closed, f is well-defined.
Let $A, B \in \mathcal{T}$, $f(A) = f(B) \implies A^c = B^c \implies A = B$. This means f is injective.
Let $C \in \mathcal{K}$. Since C is closed, it follows that C^c is open. We have $f(C^c) = (C^c)^c = C$. So every closed set is hit by f, which means that f is surjective.
Therefore, f is bijective. $\square$

Problem 2.

Let (A_n) be a nested decreasing sequence of nonempty closed sets in the metric space M.

(a) If M is complete and $\operatorname{diam} A_n \to 0$ as $n \to \infty$, show that $\bigcap A_n$ is exactly one point.

(b) To what assertions do the sets $[n, \infty)$ provide counterexamples?

Source: Real Mathematical Analysis, Pugh, Ex.52 p.129

Solution:

(a) Let $A = \bigcap A_n$. For each $n \in \mathbb{N}$, A is a subset of A_n, which implies that $\operatorname{diam} A = 0$. Since distinct points lie at a positive distance from each other, A consists of at most one point. It remains to show that A is nonempty.

Since A_n is nonempty, for each $n \in \mathbb{N}$ we can choose $a_n \in A_n$. Claim: (a_n) is a Cauchy sequence. Let $\epsilon > 0$. We can choose $N \in \mathbb{N}$ such that $\operatorname{diam} A_N < \epsilon$. It follows that for all $k, n \geq N$ we have $d_M(a_k, a_n) \leq \operatorname{diam} A_N < \epsilon$ since $a_k, a_n \in A_N$. Therefore, (a_n) is a Cauchy sequence. Since M is complete, $(a_n) \to p$, $p \in M$. For each $n \in \mathbb{N}$, the fact that A_n is closed and (a_n) lies in A_n, except possibly for finitely many terms, implies that $p \in A_n$. Thus, $p \in A$ and A is shown to be nonempty.

Therefore, $\bigcap A_n$ is exactly one point. $\square$

(b) In this part, the sequence of sets $A_n = [n, \infty)$ is missing the condition that $\operatorname{diam} A_n \to 0$ as $n \to \infty$. In fact, there is no real number x such that $x \geq n$ for all $n \in \mathbb{N}$ by the Archimedean property. This implies that $\bigcap A_n = \emptyset$. $\square$

Problem 3.

Assume that A, B are compact, disjoint, nonempty subsets of M. Prove that there are $a_0 \in A$ and $b_0 \in B$ such that for all $a \in A$ and $b \in B$ we have

$$d(a_0, b_0) \leq d(a, b).$$

[The points a_0, b_0 are closest together.]

Source: Real Mathematical Analysis, Pugh, Ex.46 p.129

Solution:

Since A, B are compact, it follows that $A \times B$ is compact. Since d is a metric, d is continuous. Therefore, $d(A \times B)$ is a compact subset of $\mathbb{R}$. This implies $d(A \times B)$ is closed and bounded. Because $d(A \times B)$ is a nonempty, closed, and bounded subset of $\mathbb{R}$, it must contain its infimum. Therefore, there exists a_0, b_0 such that $d(a_0, b_0) = \inf d(A \times B)$. $\square$

Problem 4.

Suppose that $A, B \subset \mathbb{R}^2$ are convex, closed, and have nonempty interiors. Prove that A, B are the closure of their interiors.

Source: Real Mathematical Analysis, Pugh, Ex.65 p.130

Solution:

We need to show: $A = \overline{\operatorname{int} A}$

Let $x \in \overline{\operatorname{int} A}$. Since $\operatorname{int} A \subset A \Rightarrow \overline{\operatorname{int} A} \subset \overline{A} \Rightarrow x \in \overline{A}$. A is closed $\Rightarrow A = \overline{A}$. It follows that $x \in A \Rightarrow \overline{\operatorname{int} A} \subset A$.

Now let $x \in A$. We need to show that $x \in \overline{\operatorname{int} A}$.

Since $\operatorname{int} A$ is not empty, choose $q \in \operatorname{int} A$. Consider the line segment joining x and q:

$$(l) : tx + (1-t)q, \quad 0 \leq t \leq 1$$

The claim is that any points on this line segment, possibly except x, is in the interior of A.

$$\{tx + (1-t)q \mid 0 \leq t < 1\} \subset \operatorname{int} A$$

Let $z_t = tx + (1-t)q$ for $0 \leq t < 1$.

Intuitively, we can fit a ball around z_t such that the ball is contained in A as follows:

We'll prove that it is possible to construct such a ball around z_t and that ball is contained in A.

Since $q \in \operatorname{int} A$, there is $r > 0$ such that $M_r q \subset \operatorname{int} A$, where M denotes the metric space $\mathbb{R}^2$.

Since A is convex, any convex combinations of x and points in $M_r q$ are also in A. We have:

$$tx + (1-t)M_r q = M_{(1-t)r} z_t \subset A$$

This implies that $z_t \in \operatorname{int} A$ for $0 \leq t < 1$. We can easily construct a sequence (z_n) that converges to x by simply defining z_n as:

$$z_n = \left(1 - \frac{1}{n}\right) x + \frac{1}{n} q$$

Then, $\lim z_n = x$ and (z_n) is a sequence contained in $\operatorname{int} A$. Thus, $x \in \overline{\operatorname{int} A} \Rightarrow A \subset \overline{\operatorname{int} A}$.
Therefore: $A = \overline{\operatorname{int} A}$ $\square$

Problem 5.

Suppose that M is compact and that $\mathcal{U}$ is an open covering of M which is "redundant" in the sense that each $p \in M$ is contained in at least two members of $\mathcal{U}$. Show that $\mathcal{U}$ reduces to a finite subcovering with the same property.

Source: Real Mathematical Analysis, Pugh, Ex.85 p.133

Solution:

Since M is compact and $\mathcal{U}$ is an open covering of M, $\mathcal{U}$ reduces to some V, where V is a finite subcovering. Suppose there are points in M that belong to exactly one member of V. Let's call this set of points A.
The idea is to add extra patches from $\mathcal{U}$ that are not in V to cover the points in A. For each $p \in A$, it is possible to add a set $U \in \mathcal{U}$ with $U \notin V$, such that $p \in U$, since $\mathcal{U}$ is redundant. Adding such a set for each $p \in A$ would ensure each point of M is covered by at least two sets.
If A is finite, then we only need to add a finite number of extra patches to V, and V would become a redundant finite subcovering of M.
However, it is possible that A is infinite, meaning we might need to add infinitely many extra sets to V, which makes V no longer a finite subcovering. Let's show that even if A is infinite, we only need to add a finite number of extra patches to V.
We claim that A is compact. Consider a sequence (p_n) in A. Since M is compact, (p_n) has a convergent subsequence (p_{n_k}), such that $(p_{n_k}) \to p$, where $p \in M$. Suppose $p \notin A$. This means p belongs to at least two members from V. Let V_1, V_2 be two sets that contain p. We have that $V_1 \cap V_2$ is open since a finite intersection of open sets is open. There exists $r > 0$ such that the neighborhood $M_r p \subset V_1 \cap V_2$. Since $(p_{n_k}) \to p$, there must be terms of (p_{n_k}) that lie in $M_r p$, which in turn lie in $V_1 \cap V_2$. This contradicts the fact that (p_{n_k}) is in A (because points in A belong to exactly one set).
Therefore, $p \in A$ and A is compact. Now, since A is compact, the collection of extra patches we selected to cover A is an open cover of A, and it must reduce to a finite subcovering for A.
By this construction, our original finite cover V plus this finite subcovering for A ensures that $\mathcal{U}$

reduces to a redundant finite subcovering.

Problem 6.

$(p_1, \ldots, p_n)$ is an **ϵ-chain** in a metric space M if for each i we have $p_i \in M$ and $d(p_i, p_{i+1}) < \epsilon$. The metric space is **chain-connected** if for each $\epsilon > 0$ and each pair of points $p, q \in M$ there is an ϵ-chain from p to q.

(a) Show that every connected metric space is chain-connected.

(b) Show that if M is compact and chain-connected then it is connected.

(c) Is $\mathbb{R} \setminus \mathbb{Z}$ chain-connected?

(d) If M is complete and chain-connected, is it connected?

Source: Real Mathematical Analysis, Pugh, Ex.78 p.131

Solution:

(a) Let M be a connected metric space. Suppose for contradiction that M is not chain-connected. There is an $\epsilon > 0$ and a pair of points $p, q \in M$ such that there is no ϵ-chain from p to q.

Define $A = \{x \in M \mid \text{there exists an } \epsilon\text{-chain from } p \text{ to } x\}$.

We have $p \in A$ and $q \in A^c$ so A and A^c are disjoint and nonempty.

Clearly, A is open since for any $x \in A$ we have $M_r x \subset A$ for any $r < \epsilon$. This is because having an ϵ-chain from p to x implies having an ϵ-chain from p to $y \in M_r x$, $r < \epsilon$.

Similarly, A^c is open since for any $x \in A^c$ we have $M_r x \subset A^c$ for any $r < \epsilon$. This is because if $M_r x$ is not contained in A^c then there is $y \in A \cap M_r x$. This implies there is an ϵ-chain from p to x via $y \implies x \in A$, a contradiction.

Thus, $M = A \sqcup A^c$ is a separation of M, meaning M is disconnected. This contradicts the fact that M is connected.

(b) Suppose for contradiction that M is disconnected. There is a separation of M into proper, disjoint clopen subsets:

$$M = A \sqcup A^c$$

Let $p \in A$ and $q \in A^c$. Since M is chain-connected, there exists an ϵ-chain between p and q for any $\epsilon > 0$. We construct a sequence (x_n) as follows: x_n is a point in A where $M_{1/n} x_n$ intersects both A and A^c nonempty. We are able to choose such an x_n since if for each point $x \in A$ we have $M_{1/n} x \subset A$ then it is impossible to construct a $\frac{1}{n}$-chain from p to q.

Since M is compact and A is closed, there is a subsequence $(x_{n_k}) \to c$, $c \in A$. Since A is also open, choose $\epsilon > 0$ such that $M_\epsilon c \subset A$. Because $(x_{n_k}) \to c$, there exists an $N \in \mathbb{N}$ such

that for all $n_k \geq N$ we have $x_{n_k} \in M_\epsilon c$.

Thus, it is definitely possible to find an $n_k \geq N$ such that $M_{1/n_k} x_{n_k} \subset M_\epsilon c \subset A$. However, this contradicts the fact that $M_{1/n_k} x_{n_k} \cap A^c$ is nonempty.

Therefore, M must be connected.

Intuition

An ϵ-chain must jump across the separation, and compactness forces those "jump points", which we defined by x_n, to accumulate, leading to a contradiction.

Alternative approach

Since A and A^c are compact, disjoint, nonempty subsets of M, from Problem 3, we can define the distance between A and A^c to be $\delta = d(a_0, b_0) = \inf\{d(a, b) \mid a \in A, b \in A^c\}, a_0 \in A, b_0 \in A^c$.
As $\delta > 0$, we choose ϵ such that $0 < \epsilon < \delta$. Now, it is impossible to construct an ϵ-chain from $p \in A$ to $q \in A^c$ because that would require jumping from $p_i \in A$ to $p_{i+1} \in A^c$. However, we have $d(p_i, p_{i+1}) \geq \delta > \epsilon$.

(c) $\mathbb{R} \setminus \mathbb{Z}$ is a space that is shattered into infinitely many open intervals.

Let $p, q \in \mathbb{R} \setminus \mathbb{Z}$. If p, q are in the same open interval, there exists an ϵ-chain between p and q since an open interval is connected, which implies chain-connected by part (a). If p and q are in different open intervals, it is still possible to construct an ϵ-chain from p to q. For simplicity, suppose that p and q are in two "consecutive" open intervals. Let c be the missing integer. Choose p_i to the left of c such that $d(p_i, c) < \frac{\epsilon}{2}$ and choose p_{i+1} to the right of c such that $d(p_{i+1}, c) < \frac{\epsilon}{2}$. It follows that $d(p_i, p_{i+1}) < \epsilon$.

There is an ϵ-chain from p to p_i, p_i to p_{i+1}, and p_{i+1} to q.

Therefore, $\mathbb{R} \setminus \mathbb{Z}$ is chain-connected.

Remark

Because the space is not compact, the distance between those disconnected components can be 0, which makes it possible to jump across that thin gap. Compactness would make that gap be greater than 0.

(d) Let $M = A \sqcup B$, where $A = \{(x, 0) \mid x \geq 1\}$ and $B = \{(x, 1/x) \mid x \geq 1\}$.

It is easy to verify that M is disconnected with $M = A \sqcup B$ being the separation.

Since any Cauchy sequence is bounded, it is easy to verify that M is complete.

If p and q are in the same set $\Rightarrow \exists$ an ϵ-chain from p to q.

If p and q are in different sets, we can walk as far to the right until the gap between A and B is small enough and make a jump there $\Rightarrow \exists$ an ϵ-chain from p to q.

$\Rightarrow M$ is complete, chain-connected but disconnected.

Problem 7.

A function $f : (a, b) \to \mathbb{R}$ satisfies a **Hölder condition** of order $\boldsymbol{\alpha}$ if $\alpha > 0$, and for some constant H and all $u, x \in (a, b)$ we have

$$|f(u) - f(x)| \leq H|u - x|^{\alpha}$$

The function is said to be α-Hölder, with α-Hölder constant H. (The terms "Lipschitz function of order α" and "α-Lipschitz function" are sometimes used with the same meaning.)

(a) Prove that an α-Hölder function defined on (a, b) is uniformly continuous and infer that it extends uniquely to a continuous function defined on $[a, b]$. Is the extended function α-Hölder?

(b) What does α-Hölder continuity mean when $\alpha = 1$?

(c) Prove that α-Hölder continuity when $\alpha > 1$ implies that f is constant.

Source: Real Mathematical Analysis, Pugh, Ex.2 p.198

Solution:

(a) Let $\varepsilon > 0$. Choose $\delta = \sqrt[\alpha]{\frac{\varepsilon}{H}}$. Now, for every $u, x \in (a, b)$ and $|u - x| < \delta$ implies

$$|f(u) - f(x)| \leq H|u - x|^{\alpha} < H\delta^{\alpha} = H\frac{\varepsilon}{H} = \varepsilon.$$

Thus, f is uniformly continuous on (a, b).

Remark

Uniform continuity implies that if (x_n) is Cauchy then $(f(x_n))$ is also a Cauchy sequence.

Let (x_n) be a sequence in (a, b), $(x_n) \to a$. It follows that $(f(x_n))$ also converges to a finite value L since $\mathbb{R}$ is complete. It makes sense to define $\lim_{x \to a^+} f(x) = L$ by Theorem 4.2.3 (*Understanding Analysis*, Abbott).

For g to be an extension of f that is continuous on $[a, b]$. We must define $g(a) = \lim_{x \to a^+} f(x) = L$. Likewise for endpoint b. Since this is the only way to construct such a g, g is unique.

To check if g is α-Hölder, we need to check:

(1): $|g(a) - g(x)| = |L - f(x)| \leq H|a - x|^{\alpha}$, for all $x \in (a, b)$

This is true since for any $t, x \in (a, b)$, we have:

$$\begin{aligned} & |f(t) - f(x)| \leq H|t - x|^\alpha \\ \implies & \lim_{t \to a^+} |f(t) - f(x)| \leq \lim_{t \to a^+} H|t - x|^\alpha \\ \implies & |L - f(x)| \leq H|a - x|^\alpha \end{aligned}$$

(2): $|g(b) - g(x)| \leq H|b - x|^\alpha$ (true for similar reason)

(3): $|g(b) - g(a)| \leq H|b - a|^\alpha$ (take limit of (2) as $x \to a^+$)

(b) If $\alpha = 1$, the α-Hölder condition becomes:

$$|f(u) - f(x)| \leq H|u - x|$$

Geometrically, this means the absolute value of the slope of the secant line between any two points is bounded by H.

(c) Let $\alpha > 1$. Let $u, x \in (a, b)$, $u \neq x$. We have:

$$\begin{aligned} & |f(u) - f(x)| \leq H|u - x|^\alpha \\ \implies & \left| \frac{f(u) - f(x)}{u - x} \right| \leq H|u - x|^{\alpha - 1} \\ \implies & \lim_{x \to u} \left| \frac{f(u) - f(x)}{u - x} \right| \leq \lim_{x \to u} H|u - x|^{\alpha - 1} \\ \implies & |f'(u)| \leq 0 \\ \implies & f'(u) = 0, \quad \forall u \in (a, b) \end{aligned}$$

This implies f is constant on (a, b).

Problem 8.

Assume that $f : \mathbb{R} \to \mathbb{R}$ is differentiable.

(a) If there is an $L < 1$ such that for each $x \in \mathbb{R}$ we have $f'(x) < L$, prove that there exists a unique point x such that $f(x) = x$. [x is a fixed point for f.]

(b) Show by example that (a) fails if $L = 1$.

Source: Real Mathematical Analysis, Pugh, Ex.9 p.198

Solution:

(a) We have $f'(x) < L$, $\forall x \in \mathbb{R}$. This implies that $\frac{f(a) - f(b)}{a - b} < L$, $\forall a, b \in \mathbb{R}, a < b$. Otherwise,

$\exists a, b \in \mathbb{R}, a < b : \frac{f(a)-f(b)}{a-b} \geq L$. By MVT, $\exists \theta \in (a, b) : f'(\theta) = \frac{f(a)-f(b)}{a-b} \geq L$, which is a contradiction.

Define $g(x) = f(x) - x$. We have:

$$\begin{aligned} &\frac{f(a) - f(b)}{a - b} < L \\ \implies &f(a) - f(b) > L(a - b) \\ \implies &g(a) - g(b) > (L - 1)(a - b), \quad \forall a, b \in \mathbb{R}, a < b \end{aligned}$$

Plug in $b = 0 \implies g(a) > g(0) + (L - 1)a$, $\forall a < 0$.
Since $L < 1$ and $a < 0$, we have $(L - 1)a > 0$ and this can be made arbitrarily large by letting $a \to -\infty$. This implies that $\exists a_1 < 0$ such that $g(a_1) > 0$.

Plug in $a = 0 \implies g(b) < (L - 1)b + g(0)$, $\forall b > 0$.
Similarly, we can make $g(b)$ be arbitrarily small by letting $b \to \infty$. This implies $\exists b_1 > 0 : g(b_1) < 0$.

Since g is differentiable on $\mathbb{R}$, g is also continuous and therefore has the Intermediate Value Property. There exists $\theta \in (a_1, b_1)$ such that $g(\theta) = 0$ or $f(\theta) = \theta$.

Now suppose $\exists \theta_1, \theta_2 \in \mathbb{R}$, $\theta_1 \neq \theta_2$ such that $f(\theta_1) = \theta_1$ and $f(\theta_2) = \theta_2$. It follows that:

$$1 = \frac{f(\theta_1) - f(\theta_2)}{\theta_1 - \theta_2} < L$$

which is a contradiction. Thus, the fixed point is unique.

(b) Let $L = 1$. We know that $g(x)$ is decreasing on $\mathbb{R}$ since $g'(x) = f'(x) - 1 < 0$, $\forall x \in \mathbb{R}$. We want to find an example that fails to have a root, meaning g needs to be decreasing but never cross $y = 0$.

Let $g(x) = e^{-x}$. We then have:

$$\begin{aligned} &f(x) = e^{-x} + x \\ \implies &f'(x) = 1 - e^{-x} < 1, \quad \forall x \in \mathbb{R}, \end{aligned}$$

which satisfies our desired condition.

Now, checking for a fixed point:

$$\begin{aligned} &f(x) = x \\ \implies &e^{-x} + x = x \\ \implies &e^{-x} = 0 \end{aligned}$$

But $e^{-x} > 0$, $\forall x \in \mathbb{R}$, so there is no fixed point.

Problem 9.

Suppose that $Z \subset \mathbb{R}$. Prove that the following are equivalent.

(i) Z is a zero set.

(ii) For each $\epsilon > 0$ there is a countable covering of Z by closed intervals $[a_i, b_i]$ with total length $\sum b_i - a_i < \epsilon$.

(iii) For each $\epsilon > 0$ there is a countable covering of Z by sets S_i with total diameter $\sum \operatorname{diam} S_i < \epsilon$.

Source: Real Mathematical Analysis, Pugh, Ex. 28, p.203

Solution:

(i) $\Rightarrow$ (ii): Let $\epsilon > 0$. Since Z is a zero set, there is a countable covering of Z by open intervals (a_i, b_i) such that $\sum(b_i - a_i) \leq \frac{\epsilon}{2}$. Consider the collection of closed intervals $[a_i, b_i]$. Clearly, this collection is a countable covering of Z and $\sum(b_i - a_i) \leq \frac{\epsilon}{2} < \epsilon$.
(ii) $\Rightarrow$ (iii): Trivial since we can define $S_i = [a_i, b_i]$.
(iii) $\Rightarrow$ (i): Let $\epsilon > 0$. There is a countable covering of Z by sets S_i such that $\sum \operatorname{diam} S_i < \frac{\epsilon}{2}$. Since S_i is bounded, define $a_i = \inf S_i$ and $b_i = \sup S_i$. Consider the open interval $\left(a_i - \frac{\epsilon}{2^{i+2}}, b_i + \frac{\epsilon}{2^{i+2}}\right)$. We have $S_i \subset [a_i, b_i] \subset \left(a_i - \frac{\epsilon}{2^{i+2}}, b_i + \frac{\epsilon}{2^{i+2}}\right)$, which implies this collection of open intervals is a countable covering of Z.
The total length of the covering is:

$$\sum_{i=1}^{\infty} \left(b_i - a_i + \frac{\epsilon}{2^{i+1}}\right) = \sum_{i=1}^{\infty} \operatorname{diam} S_i + \frac{\epsilon}{2} < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Appendix A

List of theorems and results